

# Ujaan Das

+44-74350-84812 | [ujaandas03@gmail.com](mailto:ujaandas03@gmail.com) | [linkedin.com/in/ujaandas](https://linkedin.com/in/ujaandas) | [github.com/ujaandas](https://github.com/ujaandas)

## EDUCATION

### University of Warwick

*Master of Science in Computer Science*

Coventry, United Kingdom

Sep. 2025 – Sep. 2026

### Hong Kong University of Science and Technology

*Bachelor of Engineering in Computer Engineering*

Hong Kong SAR

Sep. 2021 – May 2025

## EXPERIENCE

### Stellerus Technology

*Software Engineering Intern (Part Time)*

Jan. 2025 – Aug. 2025

*Hong Kong SAR*

- Re-designed a chatbot execution pipeline using a multi-step LLM workflow, implementing a custom command DSL for structured tool orchestration that reduced token overhead by 25% per session
- Standardized message contracts and schema enforcement across the pipeline, eliminating 90% of parsing-related regressions and reducing runtime errors by 30%
- Co-developed the SpaceGPT VLM platform awarded a Gold Medal at the 50th Geneva International Invention Exhibition; engineered the technical demo and automated deployment via Docker and Github Actions

### Research Assistant

*CASTLE Lab, HKUST*

May 2024 – Sep. 2024

*Hong Kong SAR*

- Engineered high-throughput semantic-analysis pipelines in Python and Java to extract semantic signals from 30k+ scraped GitHub repositories to inform research in LLM-assisted testcase generation
- Integrated Tree-sitter to parse multi-million-line codebases into ASTs and CFGs, converting complex source code into lightweight, queryable intermediate representations
- Optimized this parsing logic to cut per-project analysis time by 45%, ultimately increasing usable data yield by 3.2x to support large-scale experiments, slated for publication in ACM ICSE'27

### Stellerus Technology

*Software Engineering Intern*

May 2023 – Aug. 2023

*Hong Kong SAR*

- Developed RESTful APIs using FastAPI and MongoDB to orchestrate multi-user task queues for live satellite data, designed to sustain throughput of 8,000+ requests/sec with <150ms latency and minimal queue times
- Parallelized CPU-bottlenecked operations via a Celery/Redis worker pool, reducing processing time by 34%
- Partnered with frontend team to containerize the SPA and backend using Docker and orchestrated automated CI/CD pipelines for Azure Container Apps, cutting deployment time by 20%

## PROJECTS

### MSc Dissertation - Invariants | C++, Gtest, CMake, Python

[Github](#)

- Applying formal systems theory to LLMs for improved structured output generation via runtime constraints
- Architecting a C++ DSL and compiler with Python bindings for seamless integration into inference pipelines

### Snip | C++, Termios, Gtest, CMake, Nix

[Github](#)

- Actively maintaining a modal text editor built from ground-up with asynchronous, event-driven architecture

### yumcha | Go, C, CGo, Cocoa, Nix

[Github](#)

- Engineered a macOS tiling window manager in Go and C using Cocoa/CoreGraphics, implementing an i3-style tree layout engine for dynamic, scriptable window management without disabling SIP

### Undergraduate FYP - Follow-Me Robot | C++, Python, ROS2, Raspberry Pi, Docker

[Github](#)

- Developed a real-time UWB tracking system via custom ROS2 Python/C++ nodes, applying multi-rate Kalman and EMA filtering for stable sub-centimeter localization under noisy conditions

## SKILLS

**Languages:** C++, Python, Go, C, TypeScript, SQL, JavaScript, HTML/CSS

**Frameworks & Libraries:** GTest, CMake, FastAPI, LangGraph, Pydantic, Celery

**Databases:** MongoDB, PostgreSQL, MySQL, SQLite, Redis

**Tools & Cloud:** Docker, Nix, Azure, AWS, GitHub Actions, NGINX, Make, Git